

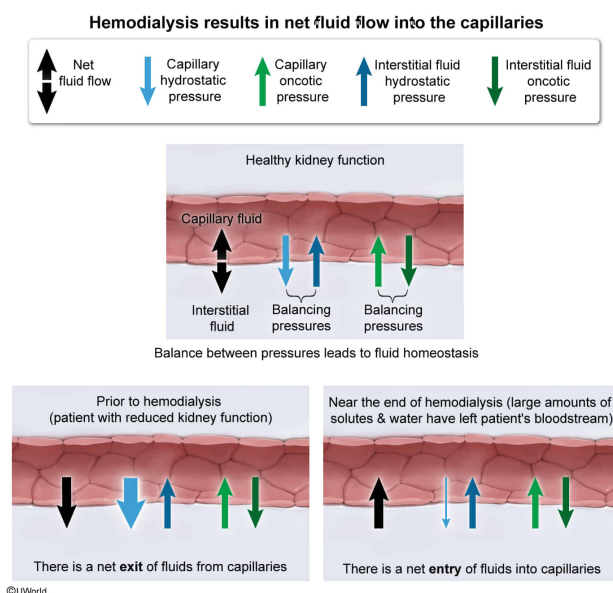
Hemodialysis is a procedure that filters blood for a patient whose kidneys are unable to do so. This procedure removes relatively large amounts of solutes and water from the patient's bloodstream in a short period of time. If a patient is undergoing hemodialysis, in which net direction would fluid flow across capillary walls near the end of the procedure?

- ☐ A. Into the capillaries, because the procedure decreased capillary solute concentration [26%]
- ✓ ☒ B. Into the capillaries, because the procedure reduced capillary hydrostatic pressure [37%]
- ☐ C. Into the interstitial fluid, because interstitial fluid osmolarity is increased [25%]
- ☐ D. Into the interstitial fluid, because overall blood volume is reduced [10%]

Correct

37% answered correctly

Explanation:



Within the cardiovascular system ([heart](#) and [blood vessels](#)), capillaries are the smallest of the blood vessels. Their small diameter and thin walls allow for the efficient exchange of gases (eg, O_2) and water (ie, via [osmosis](#)) with the interstitial fluid. The balance between [several forces](#) determines the net direction of fluid movement in capillary beds:

- Capillary hydrostatic pressure (CHP) is the pushing force of fluid (blood) against capillary walls. Increased CHP tends to cause fluids to exit capillaries, and decreased CHP tends to cause fluids to enter capillaries.

- Capillary oncotic pressure (COP) is the pulling force that solutes (proteins) in the blood exert on water (via osmosis). Increased COP tends to cause fluids to enter capillaries, and decreased COP tends to cause fluids to exit capillaries.
- Interstitial fluid hydrostatic pressure (IFHP) is the pushing force of fluid against capillary walls. Increased IFHP tends to cause fluids to enter capillaries, and decreased IFHP tends to cause fluids to exit capillaries.
- Interstitial fluid oncotic pressure (IFOP) is the pulling force that solutes in the interstitial fluid exert on blood in the capillaries. Increased IFOP tends to cause fluids to exit capillaries, and decreased IFOP tends to cause fluid entry into capillaries.

In this question, hemodialysis is described as a procedure that filters **blood** for a patient whose **kidneys** are unable to do so. Before the procedure, the patient's kidneys do not effectively filter blood, which would lead to *high* blood volume and *high* CHP, ultimately causing a net *exit* of fluids from capillaries.

The question states that hemodialysis removes relatively large amounts of solutes and water in a short period of time. As hemodialysis occurs and blood is filtered, the *removal* of solutes and fluid from the patient's body by the dialysis machine *reduces* blood volume, which would in turn *decrease* CHP. Therefore, the net direction of fluid flow near the end of the procedure is **into the capillaries** because of **reduced capillary hydrostatic pressure**.

(Choice A) Decreased capillary solute concentration tends to cause fluids to flow *out* of (*not* into) capillaries.

(Choice C) Because relatively large amounts of water and solutes are *removed* during hemodialysis, interstitial fluid **osmolarity** likely remains the same or decreases, *not* increases.

(Choice D) Reduced blood volume *decreases* CHP, causing fluid to flow into capillaries, *not* into interstitial fluid.

Educational objective:

Capillary beds function in the exchange of solutes, gases, and fluid between blood and interstitial fluid.

Time spent: 0 sec
Subject:
Biology

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System:
Circulation and Respiration